IGCSE 0607 Practice Questions — E3.4 — Combined Questions & Markscheme

Finding a quadratic function using given information

Total Questions: 27 — **Total Marks:** 145 Marks

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1 A quadratic function has x-intercepts at $x = 4$ and $x = -2$.
The coefficient of x^2 is 1.

Find the equation of the quadratic function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

2 A quadratic function $f(x)$ has x-intercepts at $x = -4$ and $x = 3$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $f(x) = x^2 + bx + c$.

$f(x) =$ ..... **[2]**

3 A quadratic function has the equation $y = x^2 + bx + c$.
The vertex of the graph of this function is $(4, -3)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

4 A quadratic function is given by

$$y = x^2 + bx + c$$

The vertex of the graph of this function is $(3, -4)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

5 A quadratic function has a minimum point at $(3, -4)$ and passes through the point $(5, 0)$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[3]**

6 The graph of a quadratic function, $y = x^2 + bx + c$, crosses the x-axis at $(-4, 0)$ and $(2, 0)$.

(a) Work out the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

7 A quadratic function $y = f(x)$ has its vertex at $(3, -4)$.
The coefficient of x^2 is 1.

(a) Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis.

..... **[2]**

8 A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

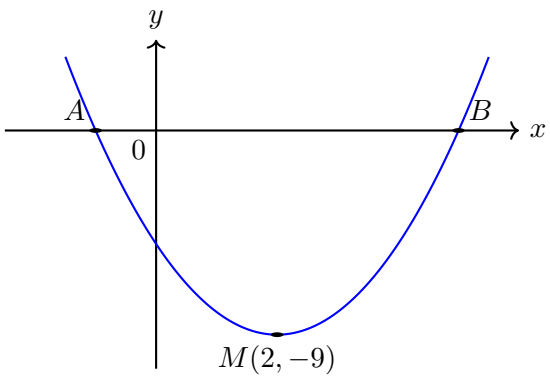
(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) The graph passes through the point $(5, p)$.
Work out the value of p.

$p =$ ..... **[2]**

9 The diagram shows the graph of a quadratic function $y = x^2 + px + q$.
The graph has a minimum point (vertex) at $M(2, -9)$.



(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) The graph crosses the x-axis at the points A and B.
Find the coordinates of A and the coordinates of B.

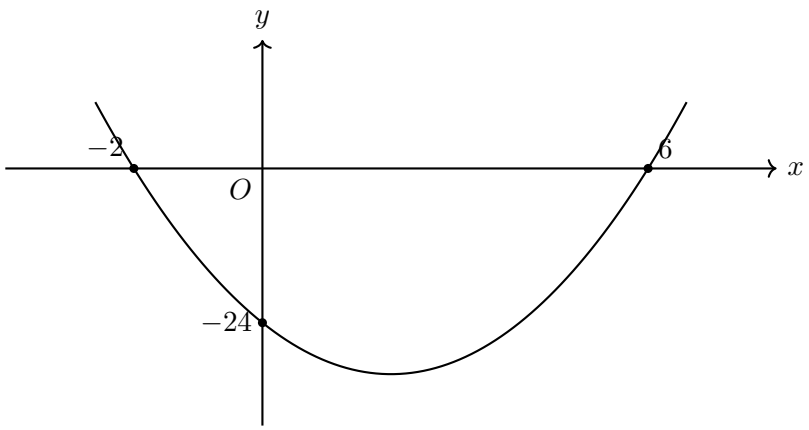
$A =$ ..... , $B =$ ..... **[3]**

10 The graph of a quadratic function $y = f(x)$ intersects the x-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the y-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

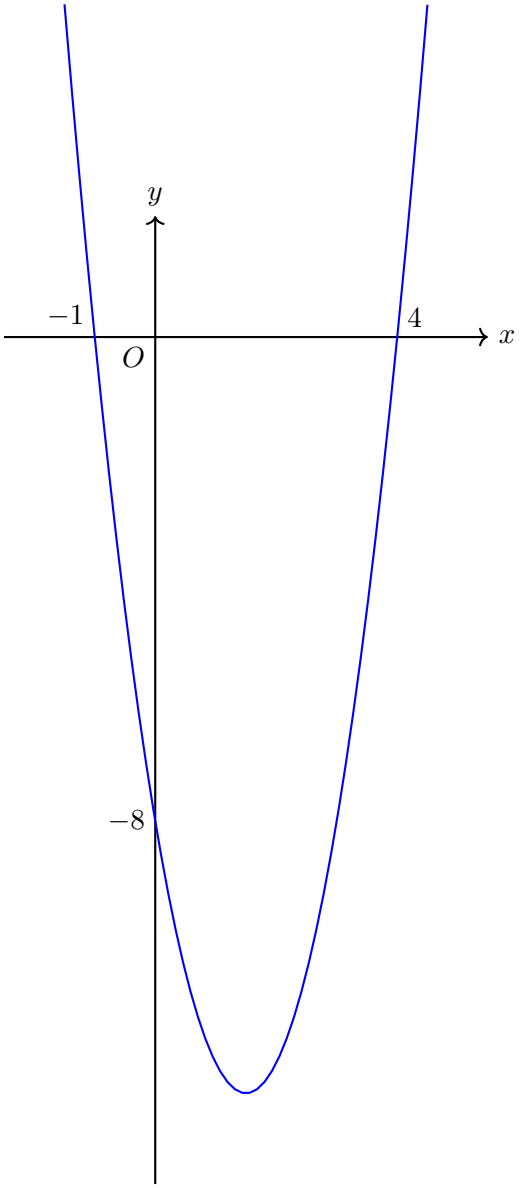
11 The diagram shows a sketch of the quadratic graph $y = f(x)$.
The graph crosses the x-axis at $(-2, 0)$ and $(6, 0)$.
It crosses the y-axis at $(0, -24)$.



Find the expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

12 The diagram shows a sketch of the quadratic function $y = f(x)$.
The graph crosses the x-axis at $(-1, 0)$ and $(4, 0)$.
The graph crosses the y-axis at $(0, -8)$.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) Find the coordinates of the minimum point of the graph.

(..... , ...) **[3]**

13 A quadratic function is given by $f(x) = a(x-h)^2 + k$.
The graph of $y = f(x)$ has a maximum point at $(-1, 8)$ and passes through the point $(2, -10)$.

(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

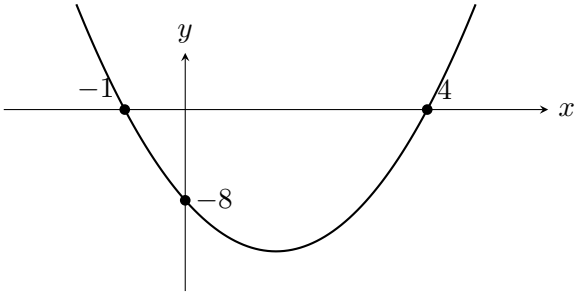
(b) Find the value of a.

$a =$ ..... **[3]**

(c) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[3]**

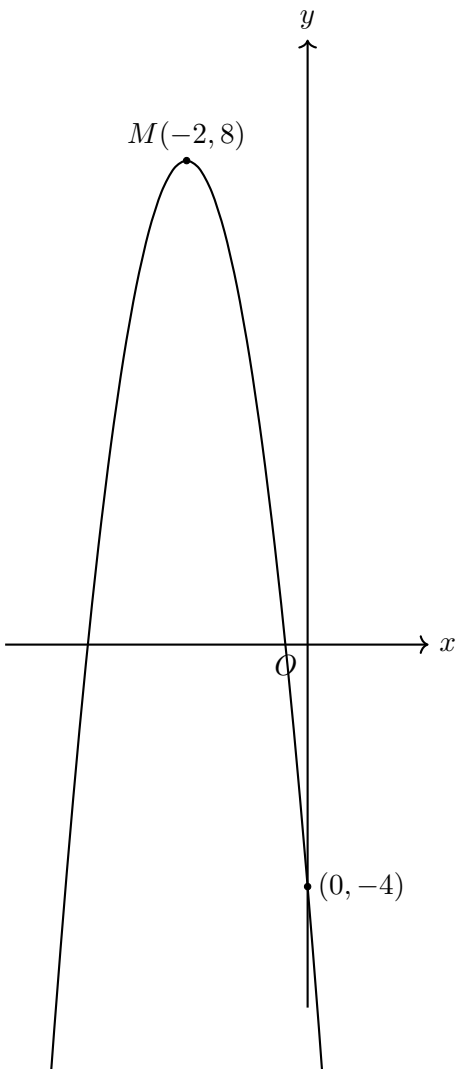
14 The diagram shows the graph of a quadratic function $y = f(x)$.
The graph crosses the x-axis at $x = -1$ and $x = 4$.
The graph crosses the y-axis at $(0, -8)$.



Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

15 The diagram shows a sketch of the curve $y = f(x)$ where $f(x)$ is a quadratic function.
The maximum point on the curve is $M(-2, 8)$ and the curve crosses the y-axis at $(0, -4)$.



(a) Find the equation of the curve.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Solve $f(x) = 0$ to find the exact x-coordinates of the points where the curve crosses the x-axis.

Give your answers in the form $\frac{p \pm q\sqrt{6}}{r}$, where p, q and r are integers.

$x =$ ..... **[3]**

16 A quadratic curve has its vertex at $(3, -5)$ and passes through the point $(1, 7)$.
Find the equation of the curve in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

17 The graph of a quadratic function has its vertex at $(2, 7)$ and passes through the point $(4, -5)$.

(a) Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

(b) Find the y-intercept of the graph.

..... **[1]**

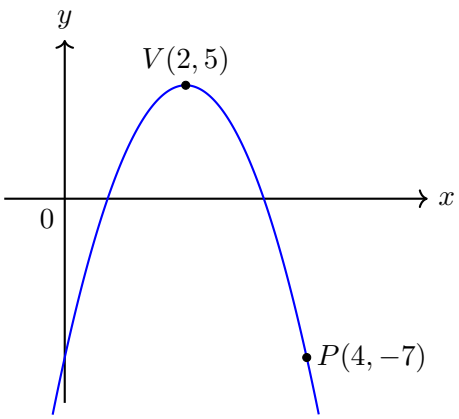
18 A quadratic curve crosses the x-axis at $(-1, 0)$ and $(5, 0)$.

The curve also passes through the point $(2, -18)$.

Find the equation of the curve in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

19 The diagram shows a sketch of the graph of a quadratic function $y = a(x - h)^2 + k$.
The vertex of the graph is $V(2, 5)$.
The graph passes through the point $P(4, -7)$.



(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

(c) Find the coordinates of the y-intercept of the graph.

(..... , ...) **[2]**

20 The graph of a quadratic function $f(x)$ intersects the x-axis at $x = -2$ and $x = 5$.
The graph passes through the point $(3, 10)$.

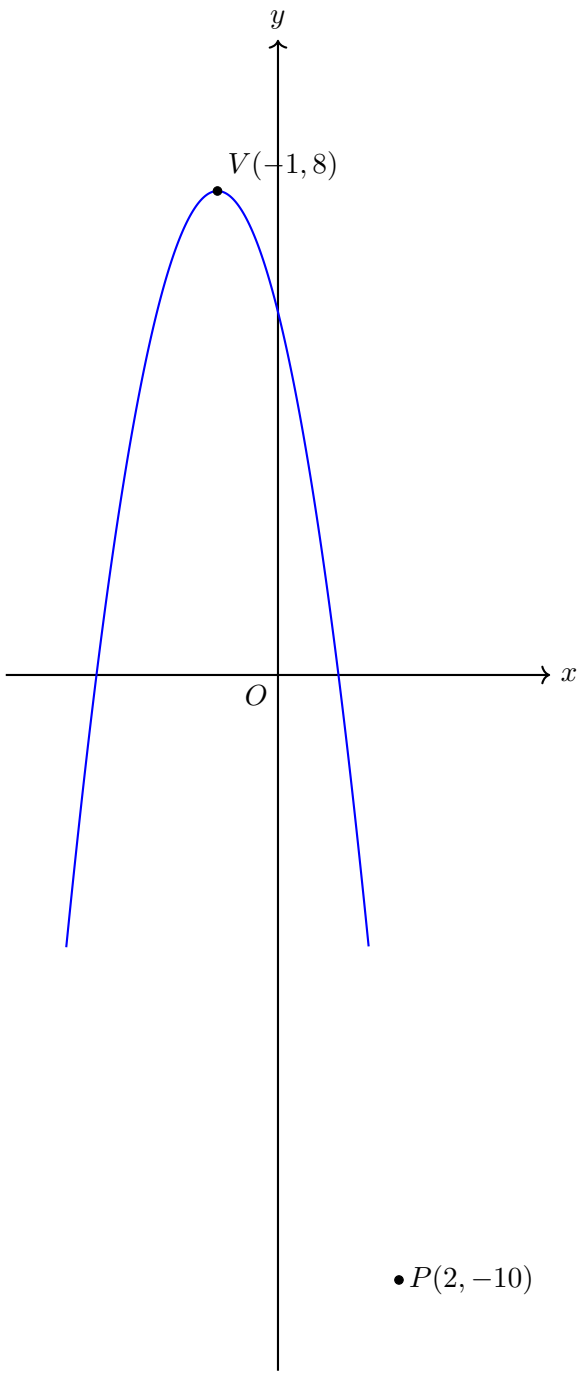
(a) Find an expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the coordinates of the vertex of the graph of $y = f(x)$.

(..... , ...) **[2]**

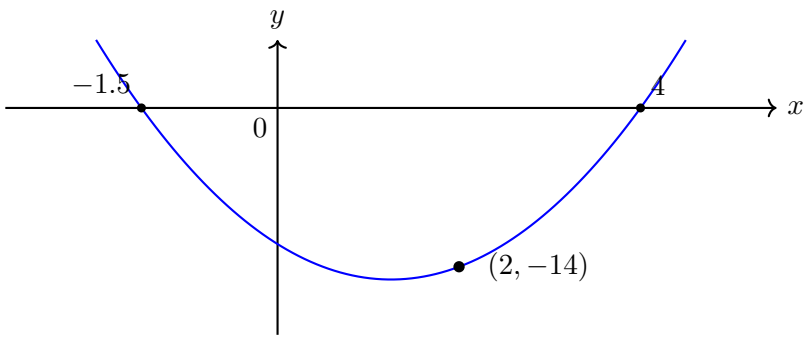
21 The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.



Work out the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

22 The diagram shows a sketch of the quadratic function $y = f(x)$.
The curve crosses the x-axis at $x = -1.5$ and $x = 4$.
It passes through the point $(2, -14)$.



(a) Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the exact coordinates of the turning point of the graph.

(..... , ...) **[3]**

23 A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The maximum point of the graph of $y = f(x)$ is $(-4, 20)$.
The graph intersects the x-axis at $x = -2$ and $x = p$.

(a) **(i)** Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(ii) Find the value of a.

$a =$ ..... **[2]**

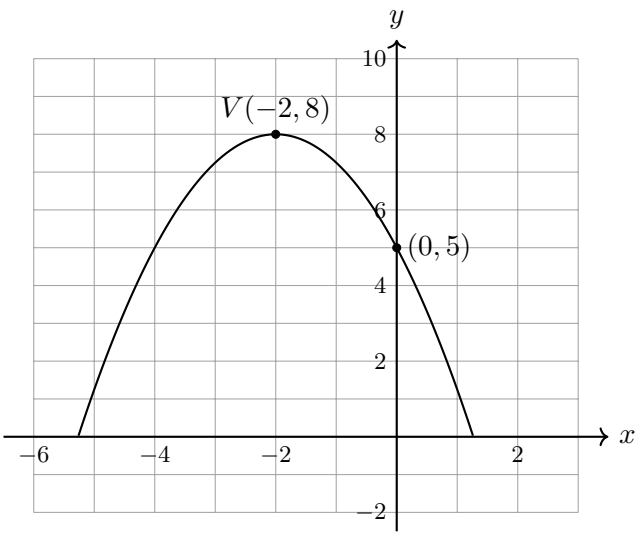
(iii) Find the value of p.

$p =$ ..... **[1]**

(iv) Express $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

24 The diagram shows the graph of a quadratic function $y = f(x)$.
The vertex of the graph is at $V(-2, 8)$ and it intersects the y-axis at $(0, 5)$.



(a) Find the equation of the function $f(x)$ in the form $f(x) = a(x - h)^2 + k$.

$f(x) =$ ..... **[3]**

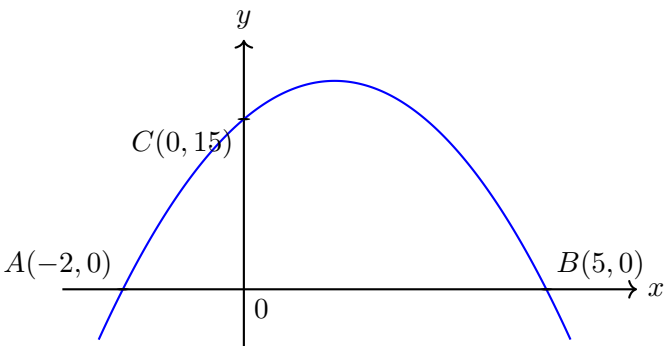
(b) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

(c) Calculate the exact x-coordinates of the points where the graph intersects the x-axis.
Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[3]**

25 The diagram shows a sketch of the quadratic curve $y = ax^2 + bx + c$.
The curve crosses the x-axis at $A(-2, 0)$ and $B(5, 0)$, and crosses the y-axis at $C(0, 15)$.



(a) Find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

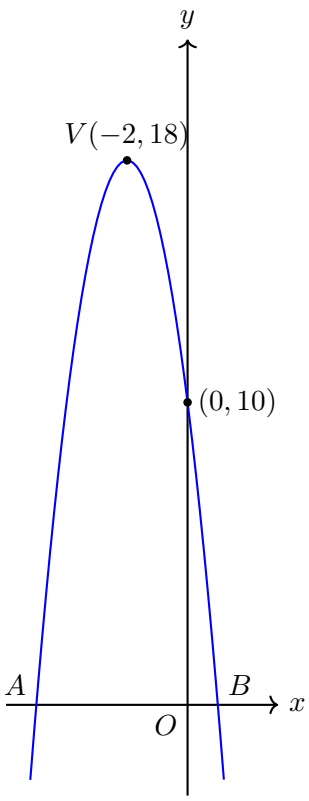
(b) Find the coordinates of the vertex of the curve.

(..... , ...) **[3]**

(c) The equation $ax^2 + bx + c = k$ has no real roots.
Write down the inequality for k.

..... **[1]**

26 The diagram shows a sketch of the graph of a quadratic function $y = f(x)$.
The maximum point of the graph is $V(-2, 18)$.
The graph crosses the y-axis at $(0, 10)$.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) The graph crosses the x-axis at A and B.
Find the length of the line segment AB.

$AB =$ ..... **[3]**

(c) The line $y = -2x + k$ is a tangent to the curve $y = f(x)$.

Find the value of k.

$k =$ ..... **[4]**

27 The quadratic function $f(x) = ax^2 + bx + c$ has a minimum turning point at $(1, -8)$.
The graph of $y = f(x)$ crosses the x-axis at $x = -1$ and $x = 3$.

(a) Show that $f(x) = 2x^2 - 4x - 6$.

[3]

(b) Solve the equation.

$f(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

(c) A straight line has equation $y = mx - 8$.
The line is a tangent to the graph of $y = f(x)$.

Find the possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

(d) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ k \end{pmatrix}$.

The new graph has no x-intercepts.

Write down the inequality for k.

..... **[2]**

1 $(x-4)(x+2)$ **[M1]**

x^2-2x-8 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Uses $(x-p)(x-q)$ with $p=-4$ and $q=3$: $(x+4)(x-3)$ **[M1]**

x^2+x-12 **[A1]**

[Back to Question] *[Total Marks: 2]*

3 Uses the vertex form $y=(x-4)^2-3$ **[M1]**

Expands to $x^2-8x+16-3$ **[M1]**

$b=-8$ and $c=13$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 Uses the vertex form $y=(x-3)^2-4$ OR uses $-\dfrac{b}{2(1)}=3$ **[M1]**

$b=-6$ **[A1]**

$c=5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

5 $y=(x-3)^2-4$ **[M1]**

$y=x^2-6x+9-4$ **[M1]**

x^2-6x+5 **[A1]**

[Back to Question] *[Total Marks: 3]*

6 (a) $x^2+bx+c=(x+4)(x-2)$ **[M1]**

$b=2$, $c=-8$ **[A1]**

(b) Finds x-coordinate of vertex: $x=\dfrac{-4+2}{2}=-1$ **[M1]**

$(-1,-9)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

7 (a) Uses vertex form: $y=(x-3)^2-4$ **[M1]**

x^2-6x+5 **[A1]**

(b) Sets $x^2-6x+5=0 \implies (x-1)(x-5)=0$ **[M1]**

$(1,0)$ and $(5,0)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** $h = 3$ **[B1]**

$k = -2$ **[B1]**

(b) $p = (5-3)^2 - 2$ **[M1]**

$p = 2$ **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** Uses vertex form: $y = (x-2)^2 - 9$ or x-coordinate of vertex $\frac{-p}{2} = 2$ **[M1]**

Expands to find p and q, or solves $2^2 + 2p + q = -9$ **[M1]**

$p = -4$ and $q = -5$ **[A1]**

(b) $x^2 - 4x - 5 = 0$ **[M1]**

$(x-5)(x+1) = 0$ **[M1]**

$A = (-1, 0)$ and $B = (5, 0)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 $f(x) = a(x+2)(x-5)$ **[M1]**

$a(0+2)(0-5) = -20$ or $-10a = -20$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 20$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 $f(x) = a(x+2)(x-6)$ **[M1]**

$-24 = a(0+2)(0-6)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 24$ **[A1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** Uses intercept form: $y = a(x+1)(x-4)$ **[M1]**

Substitutes $(0, -8)$ to find a: $-8 = a(0+1)(0-4) \implies -4a = -8 \implies a = 2$ **[M1]**

$2x^2 - 6x - 8$ **[A1]**

(b) Finds the x-coordinate of the vertex by symmetry: $x = \frac{-1+4}{2} = 1.5$ (or uses $x = -\frac{b}{2a}$) **[M1]**

Substitutes $x = 1.5$ into the equation: $y = 2(1.5)^2 - 6(1.5) - 8$ **[M1]**

$(1.5, -12.5)$ (or $(\frac{3}{2}, -\frac{25}{2})$) **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** $h = -1$ **[B1]**

$k = 8$ **[B1]**

(b) $-10 = a(2 - (-1))^2 + 8$ **[M1]**

$9a = -18$ **[M1]**

-2 **[A1]**

(c) $-2(x + 1)^2 + 8$ **[M1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$-2x^2 - 4x + 6$ **[A1]**

[Back to Question] *[Total Marks: 8]*

14 $f(x) = a(x + 1)(x - 4)$ **[M1]**

$-8 = a(0 + 1)(0 - 4)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 8$ **[A1]**

[Back to Question] *[Total Marks: 4]*

15 **(a)** Forms the equation $f(x) = a(x + 2)^2 + 8$ **[M1]**

Substitutes $(0, -4)$ into their equation:

$-4 = a(0 + 2)^2 + 8$ **[M1]**

$4a = -12 \implies a = -3$ **[A1]**

$-3x^2 - 12x - 4$ **[A1]**

(b) Sets $-3(x + 2)^2 + 8 = 0$ or $-3x^2 - 12x - 4 = 0$ **[M1]**

Rearranges to $(x + 2)^2 = \frac{8}{3}$ OR correctly substitutes into the quadratic formula:

$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-4)}}{2(-3)}$ **[M1]**

$\frac{-6 \pm 2\sqrt{6}}{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 $y = a(x - 3)^2 - 5$ **[M1]**

$7 = a(1 - 3)^2 - 5 \implies 12 = 4a \implies a = 3$ **[M1]**

$3(x - 3)^2 - 5$ **[A1]**

[Back to Question] *[Total Marks: 3]*

17 **(a)** Uses vertex form: $y = a(x-2)^2 + 7$ **[M1]**

Substitutes $(4, -5)$ into their equation: $-5 = a(4-2)^2 + 7$ **[M1]**

$-3(x-2)^2 + 7$ **[A1]**

(b) -5 **[B1]**

[Back to Question] *[Total Marks: 4]*

18 Writes the function in the factored form $y = a(x+1)(x-5)$ **[M1]**

Substitutes $(2, -18)$ into their equation:

$-18 = a(2+1)(2-5)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 10$ **[A1]**

[Back to Question] *[Total Marks: 4]*

19 **(a)** $h = 2$ **[B1]**

$k = 5$ **[B1]**

(b) Substitutes $(4, -7)$ into equation: $-7 = a(4-2)^2 + 5$ **[M1]**

$a = -3$ **[A1]**

(c) Substitutes $x = 0$ into $y = -3(x-2)^2 + 5$ **[M1]**

$(0, -7)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

20 **(a)** Sets up the factored form $f(x) = a(x+2)(x-5)$ **[M1]**

Substitutes the point $(3, 10)$: $10 = a(3+2)(3-5)$ **[M1]**

Solves for a: $10 = a(5)(-2) \implies a = -1$ **[A1]**

$f(x) = -x^2 + 3x + 10$ **[A1]**

(b) Finds the x-coordinate of the vertex (e.g., midpoint of intercepts $\frac{-2+5}{2} = 1.5$ or $x = \frac{-b}{2a}$) **[M1]**

$(1.5, 12.25)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

21 $y = a(x+1)^2 + 8$ **[M1]**

$-10 = a(2+1)^2 + 8$ **[M1]**

$9a = -18 \implies a = -2$ **[A1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$a = -2,\ b = -4,\ c = 6$ **[A1]**

[Back to Question] *[Total Marks: 5]*

22 **(a)** Sets up equation using x-intercepts: $f(x) = a(x + 1.5)(x - 4)$ OR $f(x) = a(2x + 3)(x - 4)$ **[M1]**

Substitutes $(2, -14)$ to find a: $-14 = a(2 + 1.5)(2 - 4)$ **[M1]**

Solves for a: $a = 2$ **[A1]**

$2x^2 - 5x - 12$ **[A1]**

(b) Identifies x-coordinate of vertex as midpoint of roots: $x = \dfrac{-1.5 + 4}{2} = 1.25$ OR uses $x = \dfrac{-b}{2a}$ **[M1]**

Substitutes $x = 1.25$ into their function: $y = 2(1.25)^2 - 5(1.25) - 12$ **[M1]**

$(1.25, -15.125)$ OR $\left(\dfrac{5}{4}, -\dfrac{121}{8}\right)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

23 **(a)** **(i)** $h = -4$ **[B1]**

$k = 20$ **[B1]**

(ii) $0 = a(-2 + 4)^2 + 20$ **[M1]**

$a = -5$ **[A1]**

(iii) -6 **[B1]**

(iv) $f(x) = -5(x + 4)^2 + 20 = -5(x^2 + 8x + 16) + 20$ **[M1]**

$-5x^2 - 40x - 60$ **[A1]**

[Back to Question] *[Total Marks: 7]*

24 **(a)** Uses the vertex coordinates to write $f(x) = a(x + 2)^2 + 8$ **[M1]**

Substitutes the y-intercept $(0, 5)$: $5 = a(0 + 2)^2 + 8$ **[M1]**

$f(x) = -0.75(x + 2)^2 + 8$ or equivalent using fractions like $-\dfrac{3}{4}$ **[A1]**

(b) Expands $(x + 2)^2$ as $x^2 + 4x + 4$ **[M1]**

$f(x) = -0.75x^2 - 3x + 5$ **[A1]**

(c) Equates function to zero: $-0.75(x + 2)^2 + 8 = 0$ or $-0.75x^2 - 3x + 5 = 0$ **[M1]**

Rearranges to $(x + 2)^2 = \dfrac{32}{3}$ or correctly applies the quadratic formula **[M1]**

$x = 1.27$ and $x = -5.27$ **[A1]**

[Back to Question] *[Total Marks: 8]*

25 **(a)** $c = 15$ **[B1]**

$y = a(x+2)(x-5)$ **[M1]**

$15 = a(0+2)(0-5)$ **[M1]**

$a = -1.5$ **[A1]**

$b = 4.5$ **[A1]**

(b) $x = \dfrac{-2+5}{2} = 1.5$ **[M1]**

$y = -1.5(1.5)^2 + 4.5(1.5) + 15$ **[M1]**

$(1.5, 18.375)$ or $\left(\dfrac{3}{2}, \dfrac{147}{8}\right)$ **[A1]**

(c) $k > 18.375$ **[B1]**

[Back to Question] *[Total Marks: 9]*

26 **(a)** Uses vertex form: $y = a(x+2)^2 + 18$ **[M1]**

Substitutes $(0, 10)$ to find a: $10 = a(0+2)^2 + 18 \implies 4a = -8 \implies a = -2$ **[M1]**

$-2x^2 - 8x + 10$ **[A1]**

(b) Sets $f(x) = 0$: $-2x^2 - 8x + 10 = 0$ **[M1]**

$x^2 + 4x - 5 = 0 \implies (x+5)(x-1) = 0 \implies x = -5$ or $x = 1$ **[M1]**

6 **[A1]**

(c) Equates line and curve: $-2x^2 - 8x + 10 = -2x + k$ **[M1]**

Rearranges to form a quadratic equation: $2x^2 + 6x + (k-10) = 0$ (or equivalent) **[M1]**

Sets discriminant $\Delta = 0$ for a tangent: $6^2 - 4(2)(k-10) = 0$ **[M1]**

14.5 (or $\dfrac{29}{2}$) **[A1]**

[Back to Question] *[Total Marks: 10]*

27 **(a)** Forms equation using x-intercepts: $f(x) = a(x+1)(x-3)$ OR using vertex $f(x) = a(x-1)^2 - 8$ **[M1]**

Substitutes appropriate point to find a: $a(1+1)(1-3) = -8 \implies -4a = -8 \implies a = 2$ **[M1]**

Expands to show final form: $2(x^2 - 2x - 3) = 2x^2 - 4x - 6$ **[A1]**

(b) $2x^2 - 4x - 6 = 10 \implies 2x^2 - 4x - 16 = 0$ **[M1]**

Factors correctly to $(x-4)(x+2) = 0$ OR uses quadratic formula correctly **[M1]**

$x = 4$ and $x = -2$ **[A1]**

(c) Sets equations equal: $2x^2 - 4x - 6 = mx - 8$ leading to $2x^2 - (4+m)x + 2 = 0$ **[M1]**

States condition for tangent (discriminant $= 0$): $(-(4+m))^2 - 4(2)(2) = 0$ **[M1]**

Simplifies to $(4+m)^2 = 16$ or $m^2 + 8m = 0$ **[M1]**

$m = 0$, $m = -8$ **[A1]**

(d) Recognises minimum y-value must be strictly greater than 0: $-8 + k > 0$ **[M1]**

$k > 8$ **[A1]**

[Back to Question] *[Total Marks: 12]*